

Name: Krunal

Homework assignment 3: Rock, Paper Scissors, Lizard, Spock

Assumption:

1. In the game server will work as the central administrator and 5 clients (Players) will join the game.
2. To start the game, all 5 players have to join 1st.
3. After joining the game, the player can quit the game.
4. There will be 5 rounds for the game, and after having 5 rounds the winner will be declared.
5. After starting the game, players can quit but no other player can join the middle of the game.

How to run code?

1. Open terminal.
2. Clone the repository.
3. Go to the HW3 folder.
4. Run the makefile
 - a. cmd: make clean && make
 - b. ./spock -s 51515
(Note: You can select any port(free) as per your convenience)
This terminal will act as the server.
5. Open other 5 terminals
6. Run client mode
 - a. cmd: ./spock -c 51515
7. Enjoy playing the game!!.

Screenshots of the output:

Server Terminal Logging:

```
krunal@Mac HW3 % ./spock -s 51515
Server is running on port 51515
Server has accepted connection from player index at 0. Now Total Active Players: 1
Krunal (Player 1) joined! at Addr: 127.0.0.1 Port: 57438
Krunal is waiting to start game.!!
Server has accepted connection from player index at 1. Now Total Active Players: 2
Axy (Player 2) joined! at Addr: 127.0.0.1 Port: 57439
Axy is waiting to start game.!!
Server has accepted connection from player index at 2. Now Total Active Players: 3
Hetav (Player 3) joined! at Addr: 127.0.0.1 Port: 57440
Hetav is waiting to start game.!!
Server has accepted connection from player index at 3. Now Total Active Players: 4
krutik (Player 4) joined! at Addr: 127.0.0.1 Port: 57441
krutik is waiting to start game.!!
Server has accepted connection from player index at 4. Now Total Active Players: 5
Maximum players count reached. Lets start the game loop...
Broadcasting game start signal...
Game loop thread has been started.
Axy notified about start game!
Hetav notified about start game!
krutik notified about start game!
Game loop: round 1 started. Waiting for 5 players...
Krunal notified about start game!
```

Adit (Player 5) joined! at Addr: 127.0.0.1 Port: 57443
Adit is waiting to start game.!!
Adit notified about start game!
Krunal has played move R for 1 round.
Krunal is waiting for results of round 1 ...!
Axay has played move P for 1 round.
Axay is waiting for results of round 1 ...!
Hetav has played move S for 1 round.
Hetav is waiting for results of round 1 ...!
krutik has played move L for 1 round.
krutik is waiting for results of round 1 ...!
Adit has played move K for 1 round.
Adit is waiting for results of round 1 ...!
Game loop: All 5 players are ready for playing the round 1. Processing...
Krunal (Rock) won 2 matchups. Adding 2 to final score. (Prev score: 0)
Axay (Paper) won 2 matchups. Adding 2 to final score. (Prev score: 0)
Hetav (Scissors) won 2 matchups. Adding 2 to final score. (Prev score: 0)
krutik (Lizard) won 2 matchups. Adding 2 to the final score. (Prev score: 0)
Adit (Spock) won 2 matchups. Adding 2 to final score. (Prev score: 0)
Game loop: Round 1 has been finished.
Game loop: round 2 started. Waiting for 5 players...
Krunal moving forward after completing round 1.
Axay moving forward after completing round 1.
Hetav moving forward after completing round 1.
krutik moving forward after completing round 1.
Adit moving forward after completing round 1.
Krunal has played move R for 2 round.
Krunal is waiting for results of round 2 ...!
Axay has played move R for 2 round.
Axay is waiting for results of round 2 ...!
Hetav has played move R for 2 round.
Hetav is waiting for results of round 2 ...!
krutik has played move S for 2 round.
krutik is waiting for results of round 2 ...!
Adit has played move S for 2 round.
Adit is waiting for results of round 2 ...!
Game loop: All 5 players are ready for playing the round 2. Processing...
Krunal (Rock) won 2 matchups. Adding 2 to final score. (Prev score: 2)
Axay (Rock) won 2 matchups. Adding 2 to final score. (Prev score: 2)
Hetav (Rock) won 2 matchups. Adding 2 to final score. (Prev score: 2)
Game loop: Round 2 has been finished.
Game loop: round 3 started. Waiting for 5 players...
Axay moving forward after completing round 2.
Hetav moving forward after completing round 2.
krutik moving forward after completing round 2.
Adit moving forward after completing round 2.
Krunal moving forward after completing round 2.
Krunal has played move L for 3 round.
Krunal is waiting for results of round 3 ...!
Axay has played move L for 3 round.
Axay is waiting for results of round 3 ...!
Hetav has played move L for 3 round.
Hetav is waiting for results of round 3 ...!
krutik has played move L for 3 round.
krutik is waiting for results of round 3 ...!
Adit has played move L for 3 round.
Adit is waiting for results of round 3 ...!
Game loop: All 5 players are ready for playing the round 3. Processing...
Game loop: Round 3 has been finished.
Game loop: round 4 started. Waiting for 5 players...
Krunal moving forward after completing round 3.
Axay moving forward after completing round 3.
Hetav moving forward after completing round 3.
krutik moving forward after completing round 3.
Adit moving forward after completing round 3.
Krunal has requested to reset score values.
Krunal has played move R for 4 round.
Krunal is waiting for results of round 4 ...!

Axay has played move R for 4 round.
Axay is waiting for results of round 4 ...!
Hetav has played move R for 4 round.
Hetav is waiting for results of round 4 ...!
krutik has played move P for 4 round.
krutik is waiting for results of round 4 ...!
Adit has played move P for 4 round.
Adit is waiting for results of round 4 ...!
Game loop: All 5 players are ready for playing the round 4. Processing...
krutik (Paper) won 3 matchups. Adding 3 to final score. (Prev score: 0)
Adit (Paper) won 3 matchups. Adding 3 to final score. (Prev score: 0)
Game loop: Round 4 has been finished.
Game loop: round 5 started. Waiting for 5 players...
Krunal moving forward after completing round 4.
Axay moving forward after completing round 4.
Hetav moving forward after completing round 4.
krutik moving forward after completing round 4.
Adit moving forward after completing round 4.
Krunal has played move R for 5 round.
Krunal is waiting for results of round 5 ...!
Axay has played move P for 5 round.
Axay is waiting for results of round 5 ...!
Hetav has played move S for 5 round.
Hetav is waiting for results of round 5 ...!
krutik has played move L for 5 round.
krutik is waiting for results of round 5 ...!
Adit has played move K for 5 round.
Adit is waiting for results of round 5 ...!
Game loop: All 5 players are ready for playing the round 5. Processing...
Krunal (Rock) won 2 matchups. Adding 2 to final score. (Prev score: 0)
Axay (Paper) won 2 matchups. Adding 2 to final score. (Prev score: 0)
Hetav (Scissors) won 2 matchups. Adding 2 to final score. (Prev score: 0)
krutik (Lizard) won 2 matchups. Adding 2 to final score. (Prev score: 3)
Adit (Spock) won 2 matchups. Adding 2 to final score. (Prev score: 3)
Game loop: Round 5 has been finished.
Game loop: all rounds done.
Game loop thread has been finished.
Showing final scores...
Final Score - Krunal: 2
Final Score - Axay: 2
Final Score - Hetav: 2
Final Score - krutik: 5
Final Score - Adit: 5
Broadcasting final results everywhere:
===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axay: 2
Hetav: 2
krutik: 5
Adit: 5
Winners (Tie game between): krutik, Adit with 5 points!
Krunal moving forward after completing round 5.
Thread for Krunal (idx: 0) has been cleaned up!!
Decrementing count for active players now: 4
Axay moving forward after completing round 5.
Thread for Axay (idx: 1) has been cleaned up!!
Decrementing count for active players now: 3
Hetav moving forward after completing round 5.
Thread for Hetav (idx: 2) has been cleaned up!!
Decrementing count for active players now: 2
krutik moving forward after completing round 5.
Thread for krutik (idx: 3) has been cleaned up!!
Decrementing count for active players now: 1
Adit moving forward after completing round 5.
Thread for Adit (idx: 4) has been cleaned up!!
Decrementing count for active players now: 0

Client Terminal 1: Player 1(Krunal)

```
[krunal@Mac HW3 % ./spock -c 51515
Enter your name: Krunal
Connected as the name Krunal. Waiting for server to give response...
4 players remaining to join!
(M=Score, Q=Quit, T=Reset Scores)

Round 1/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 1 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 2)
Adit: Spock (Current_Score: 2)

Round 2/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 2 ===
Krunal: Rock (Current_Score: 4)
Axy: Rock (Current_Score: 4)
Hetav: Rock (Current_Score: 4)
krutik: Scissors (Current_Score: 2)
Adit: Scissors (Current_Score: 2)

Round 3/5
Choose (R/P/S/L/K/Q/M/T): Your choice:
No input has been entered. Please choose R/P/S/L/K/Q/M/T.
Your choice: D
Invalid input 'D'. Please choose R/P/S/L/K/Q/M/T.
Your choice: L

=== RESULTS AFTER ROUND 3 ===
Krunal: Lizard (Current_Score: 4)
Axy: Lizard (Current_Score: 4)
Hetav: Lizard (Current_Score: 4)
krutik: Lizard (Current_Score: 2)
Adit: Lizard (Current_Score: 2)

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: M
Your current game score is : 4

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: T
score values have been reset by a player.

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 4 ===
Krunal: Rock (Current_Score: 0)
Axy: Rock (Current_Score: 0)
Hetav: Rock (Current_Score: 0)
krutik: Paper (Current_Score: 3)
Adit: Paper (Current_Score: 3)

Round 5/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 5 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 5)
Adit: Spock (Current_Score: 5)

===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axy: 2
Hetav: 2
krutik: 5
Adit: 5

Winners (Tie game between): krutik, Adit with 5 points!

Server is disconnected.
Connection is closed.
krunal@Mac HW3 %
```

In Round 3 error Handling has been shown and before starting the Round 4,

'M' has been entered, which shows the score of the player at the end of the previous round.

'T' has been entered, resetting each player's score to 0. (Note: Who reset the score, this will be shown in server terminal only, in client terminals player will be notified that someone(player) reset the score.

Client Terminal 2: Player 2(Axay)

```
krunal@Mac HW3 % ./spock -c 51515
Enter your name: Axay
Connected as the name Axay. Waiting for server to give response...
3 players remaining to join!
(M=Score, Q=Quit, T=Reset Scores)

Round 1/5
Choose (R/P/S/L/K/Q/M/T): Your choice: P

=== RESULTS AFTER ROUND 1 ===
Krunal: Rock (Current_Score: 2)
Axay: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 2)
Adit: Spock (Current_Score: 2)

Round 2/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 2 ===
Krunal: Rock (Current_Score: 4)
Axay: Rock (Current_Score: 4)
Hetav: Rock (Current_Score: 4)
krutik: Scissors (Current_Score: 2)
Adit: Scissors (Current_Score: 2)

Round 3/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 3 ===
Krunal: Lizard (Current_Score: 4)
Axay: Lizard (Current_Score: 4)
Hetav: Lizard (Current_Score: 4)
krutik: Lizard (Current_Score: 2)
Adit: Lizard (Current_Score: 2)

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R
score values have been reset by a player.

=== RESULTS AFTER ROUND 4 ===
Krunal: Rock (Current_Score: 0)
Axay: Rock (Current_Score: 0)
Hetav: Rock (Current_Score: 0)
krutik: Paper (Current_Score: 3)
Adit: Paper (Current_Score: 3)

Round 5/5
Choose (R/P/S/L/K/Q/M/T): Your choice: P

=== RESULTS AFTER ROUND 5 ===
Krunal: Rock (Current_Score: 2)
Axay: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 5)
Adit: Spock (Current_Score: 5)

===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axay: 2
Hetav: 2
krutik: 5
Adit: 5

Winners (Tie game between): krutik, Adit with 5 points!

Server is disconnected.
Connection is closed.
krunal@Mac HW3 %
```

Client 3: Player 3(Hetav)

```
krunal@Mac HW3 % CLEAR

krunal@Mac HW3 % ./spock -c 51515
Enter your name: Hetav
Connected as the name Hetav. Waiting for server to give response...
2 players remaining to join!
(M=Score, Q=Quit, T=Reset Scores)

Round 1/5
Choose (R/P/S/L/K/Q/M/T): Your choice: S

=== RESULTS AFTER ROUND 1 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 2)
Adit: Spock (Current_Score: 2)

Round 2/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R

=== RESULTS AFTER ROUND 2 ===
Krunal: Rock (Current_Score: 4)
Axy: Rock (Current_Score: 4)
Hetav: Rock (Current_Score: 4)
krutik: Scissors (Current_Score: 2)
Adit: Scissors (Current_Score: 2)

Round 3/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 3 ===
Krunal: Lizard (Current_Score: 4)
Axy: Lizard (Current_Score: 4)
Hetav: Lizard (Current_Score: 4)
krutik: Lizard (Current_Score: 2)
Adit: Lizard (Current_Score: 2)

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: R
score values have been reset by a player.

=== RESULTS AFTER ROUND 4 ===
Krunal: Rock (Current_Score: 0)
Axy: Rock (Current_Score: 0)
Hetav: Rock (Current_Score: 0)
krutik: Paper (Current_Score: 3)
Adit: Paper (Current_Score: 3)

Round 5/5
Choose (R/P/S/L/K/Q/M/T): Your choice: S

=== RESULTS AFTER ROUND 5 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 5)
Adit: Spock (Current_Score: 5)

===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axy: 2
Hetav: 2
krutik: 5
Adit: 5

Winners (Tie game between): krutik, Adit with 5 points!

Server is disconnected.
Connection is closed.
krunal@Mac HW3 %
```

Client 4: Player 4(Krutik)

```
HW3 — zsh — 101x71

krunal@Mac HW3 % ./spock -c 51515
Enter your name: krutik
Connected as the name krutik. Waiting for server to give response...
1 players remaining to join!
(M=Score, Q=Quit, T=Reset Scores)

Round 1/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 1 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 2)
Adit: Spock (Current_Score: 2)

Round 2/5
Choose (R/P/S/L/K/Q/M/T): Your choice: S

=== RESULTS AFTER ROUND 2 ===
Krunal: Rock (Current_Score: 4)
Axy: Rock (Current_Score: 4)
Hetav: Rock (Current_Score: 4)
krutik: Scissors (Current_Score: 2)
Adit: Scissors (Current_Score: 2)

Round 3/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 3 ===
Krunal: Lizard (Current_Score: 4)
Axy: Lizard (Current_Score: 4)
Hetav: Lizard (Current_Score: 4)
krutik: Lizard (Current_Score: 2)
Adit: Lizard (Current_Score: 2)

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: P
score values have been reset by a player.

=== RESULTS AFTER ROUND 4 ===
Krunal: Rock (Current_Score: 0)
Axy: Rock (Current_Score: 0)
Hetav: Rock (Current_Score: 0)
krutik: Paper (Current_Score: 3)
Adit: Paper (Current_Score: 3)

Round 5/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 5 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 5)
Adit: Spock (Current_Score: 5)

===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axy: 2
Hetav: 2
krutik: 5
Adit: 5

Winners (Tie game between): krutik, Adit with 5 points!

Server is disconnected.
Connection is closed.
krunal@Mac HW3 %
```


Client 5: Player 5(Adit)

```
HW3 — -zsh — 100x72
[krunal@Mac HW3 % CLEAR

[krunal@Mac HW3 % ./spock -c 51515
Enter your name: Adit
Connected as the name Adit. Waiting for server to give response...
All players joined the GAME! Let's start the game...!
(M=Score, Q=Quit, T=Reset Scores)

Round 1/5
Choose (R/P/S/L/K/Q/M/T): Your choice: K

=== RESULTS AFTER ROUND 1 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 2)
Adit: Spock (Current_Score: 2)

Round 2/5
Choose (R/P/S/L/K/Q/M/T): Your choice: S

=== RESULTS AFTER ROUND 2 ===
Krunal: Rock (Current_Score: 4)
Axy: Rock (Current_Score: 4)
Hetav: Rock (Current_Score: 4)
krutik: Scissors (Current_Score: 2)
Adit: Scissors (Current_Score: 2)

Round 3/5
Choose (R/P/S/L/K/Q/M/T): Your choice: L

=== RESULTS AFTER ROUND 3 ===
Krunal: Lizard (Current_Score: 4)
Axy: Lizard (Current_Score: 4)
Hetav: Lizard (Current_Score: 4)
krutik: Lizard (Current_Score: 2)
Adit: Lizard (Current_Score: 2)

Round 4/5
Choose (R/P/S/L/K/Q/M/T): Your choice: P
score values have been reset by a player.

=== RESULTS AFTER ROUND 4 ===
Krunal: Rock (Current_Score: 0)
Axy: Rock (Current_Score: 0)
Hetav: Rock (Current_Score: 0)
krutik: Paper (Current_Score: 3)
Adit: Paper (Current_Score: 3)

Round 5/5
Choose (R/P/S/L/K/Q/M/T): Your choice: K

=== RESULTS AFTER ROUND 5 ===
Krunal: Rock (Current_Score: 2)
Axy: Paper (Current_Score: 2)
Hetav: Scissors (Current_Score: 2)
krutik: Lizard (Current_Score: 5)
Adit: Spock (Current_Score: 5)

===== GAME IS OVER =====
FINAL SCORES ARE :
Krunal: 2
Axy: 2
Hetav: 2
krutik: 5
Adit: 5

Winners (Tie game between): krutik, Adit with 5 points!

Server is disconnected.
Connection is closed.
[krunal@Mac HW3 % █
```


Examples of game:

1.

Krunal: R

Hetav: R

Axay: R

Krutik: P

Adit: P

Score determination:

Krutik(P) wins over Krunal(+1),Hetav(+1),Axay(+1) = Total 3

Adit(P) wins over Krunal(+1),Hetav(+1),Axay(+1) = Total 3

Same kind of moves and losing match won't affect the score so Krunal, Hetav, Axay will have same score which will be '0'

2.

Krunal: R

Hetav: R

Axay: R

Krutik: R

Adit: R

Everyone will get 0 as the score value at the end of the round.

3.

Krunal: R

Hetav: P

Axay: S

Krutik: L

Adit: K

Krutik(L) wins over Adit(K) and Hetav(P) = Total 2

Adit(K) wins over Axay(S) and Krunal(R) = Total 2

Axay(S) wins over Hetav(P) and Krutik(L) = Total 2

Krunal(R) wins over Krutik(L) and Axay(S) = Total 2

Hetav(P) wins over Krunal(R) and Adit(K) = Total 2

Last Scenario:

Let's say 5 players joined in the start and assume Krunal having the score 5 at the end of the 2nd round.

Krunal then quit the game.

Then 4 people will continue to play,

At the last when showing the result

Score of Krunal also will be shown.

Github Repo Link:

https://github.khoury.northeastern.edu/klsavaj/CS5700_klsavaj/tree/main/HW3